

Tabassum Haque Rose

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RESEARCH INTERESTS

My research interests lie at the intersection of artificial intelligence, machine learning, and deep learning, spanning image processing, pattern recognition, representation learning, and medical imaging. I am particularly interested in developing robust and generalizable models for image analysis, recognition, and predictive tasks, with applications in healthcare and other human-centered domains. The through-line across my interests is developing reliable explainable AI and AI-driven healthcare systems that can learn meaningful representations from complex visual data and support real-world applications.

EDUCATION

B.Sc. in Computer Science & Engineering

2017 – 2023

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

Thesis: *Human Emotion Recognition from Facial Expressions Using CNNs and Transfer Learning*

Relevant coursework: Digital Image Processing · Artificial Intelligence · Digital Signal Processing · Neural Networks & Fuzzy Systems · Data Mining · Network Security · Database Systems

RESEARCH EXPERIENCE

Undergraduate Thesis — Facial Expression-Based Emotion Recognition

Department of Computer Science & Engineering, RUET

- As the sole contributor, carried out the core research pipeline for facial emotion recognition, using JAFFE, FER-2013, and CK+ datasets and experimenting with four transfer learning architectures (VGG16, ResNet152V2, DenseNet169, MobileNet) and a custom CNN for seven-class emotion classification.
- Conducted data preprocessing through image resizing and augmentation, and applied transfer learning with frozen early layers, dropout, batch normalization, L2 regularization, and softmax activation.
- Benchmarked model performance across three datasets using accuracy, precision, recall, F1-score, AUC, and confusion matrices, with MobileNet achieving the strongest classification results.

Current Works in Progress

- Rose, Tabassum**, et al. “Pixel to Patterns: A Comprehensive Review of Deep Learning and NLP in Neuroradiology.”
- Rose, Tabassum**, et al. “From Radical-Pair Magnetoreception to Engineered Sensing: A Scoping Review of Quantum Biology, Translation, and Artificial Intelligence.”

PROFESSIONAL EXPERIENCE

Linguistic AI Specialist(Bangla) — Outlier AI (Remote & Part-time)

Jan 2025– June 2025

- Analyzed & Evaluated language and audio tasks in the bangla language for AI model training, analyzing model outputs for accuracy, relevance, and linguistic quality and consistency.
- Performed structured data annotation and response refinement to support improvement of AI-generated outputs.

Part-Time STEM Instructor — Schools in Bangladesh

Part-time

- Taught mathematics and computer programming to school students, adapting technical concepts and lessons to different grade levels and learning needs.
- Mentored students in programming fundamentals and computational problem-solving through hands-on projects and prepared teams for school-level IT based competitions.

TECHNICAL SKILLS

Programming: Python · C · C++ · SQL · Dart

Machine learning & Deep learning: · TensorFlow · Keras · OpenCV · Scikit-learn · CNN · Transfer learning · Predictive modeling · Neural networks

Image Processing & Analysis: Image Preprocessing · Data Augmentation · Feature Extraction · Pattern recognition

Data Analysis & Visualization: Numpy · Pandas · Matplotlib · Seaborn · Exploratory data analysis

Tools: Git & Github · VS Code · Anaconda · Google Colab · MS Office · Kaggle · LaTeX

Soft Skills: Presentation · Technical Writing · Project Management

SELECTED PROJECTS

- **Age & Gender Prediction using CNN** — Implemented a CNN-based approach for simultaneous age estimation and gender classification using the UTKFace dataset, applying image preprocessing, normalization, and dropout regularization.
- **Breast Cancer Prediction** — Recreated a machine-learning approach for breast cancer diagnosis using standardized clinical features and Logistic Regression, with exploratory analysis and train/test evaluation.
- **Image Classification for Industrial Equipment** — Applied and evaluated a CNN-based approach to industrial image classification, using image augmentation, batch normalization, dropout, and validation-based training techniques.
- **Credit Card Fraud Detection** — Utilized Logistic Regression, Random Forest, and XGBoost for fraud classification, using feature scaling and exploratory analysis; addressed class imbalance with SMOTE and evaluated models using F1-score.

AWARDS & LEADERSHIP

- **National Education Board Merit Scholarship (Higher Secondary School Certificate)** — By The Government of the People's Republic of Bangladesh (2017-2022).
- **National Education Board Merit Scholarship (Secondary School Certificate)** — By The Government of the People's Republic of Bangladesh (2014-2016).
- **HULT PRIZE** — Coordinator of Event Management, RUET (2019).
- **Student to Startup: Chapter 2(Bangladesh's National Student Entrepreneurship Competition)** — Campus Ambassador (2019)
- **UNIBATOR(Joint Innovation and Entrepreneurship Initiative)** — Campus Ambassador (2021)
- **RUET Students Association of Saidpur** — Joint General Secretary, RUET (2021–22).

CERTIFICATIONS

- **Machine Learning with Python: Foundations** — Linkedin Learning, 2025.
- **Artificial Intelligence Foundations: Machine Learning** — Linkedin Learning, 2025.
- **Python for Deep Learning: Build Neural Network in Python** — Udemy, 2025.
- **Learning Data Science: Understanding the basics** — Linkedin Learning, 2025.
- **Artificial Intelligence in Healthcare** — Udemy, 2026.
- **Career Essentials in CyberSecurity** — Microsoft and Linkedin Learning, 2025.

REFERENCES

Dr. Md. Al Mamun — Professor(Thesis Supervisor), Department of Computer Science & Engineering, RUET, Rajshahi-6204, Bangladesh
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Dr. Mir Md. Jahangir Kabir — Professor, Department of Computer Science & Engineering, RUET, Rajshahi-6204, Bangladesh
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